

Engineering Mathematics: Applying Euler's Theorem

Here are the step-by-step solutions for the two practical problems from the lecture notes. These examples showcase how to quickly solve partial differential equations using Euler's Theorem by finding the degree n of a homogeneous function.

Question 1

Problem Statement

If $u = \frac{x+y}{x^2+y^2}$, find the value of:

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$$

Step-by-Step Solution

Define the function explicitly:

$$f(x, y) = \frac{x + y}{x^2 + y^2}$$

Substitute $x \rightarrow xt$ and $y \rightarrow yt$ to verify homogeneity and extract the degree:

$$f(xt, yt) = \frac{(xt) + (yt)}{(xt)^2 + (yt)^2}$$

Factor out the scaling parameter t from the numerator and denominator:

$$f(xt, yt) = \frac{t(x + y)}{t^2(x^2 + y^2)}$$

Simplify the powers of t using the exponent rule $\frac{t^1}{t^2} = t^{1-2} = t^{-1}$:

$$f(xt, yt) = t^{-1} \left(\frac{x + y}{x^2 + y^2} \right)$$

Substitute the original definition of u back into the equation:

$$f(xt, yt) = t^{-1}u$$

This proves that u is a homogeneous function with a degree of $n = -1$.

By applying **Euler's Theorem** ($x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$):

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -1 \cdot u = -u$$

Final Answer

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -\frac{x + y}{x^2 + y^2}$$

Question 2

Problem Statement

If $u = \frac{\sqrt{x} + \sqrt{y}}{x + y}$, find the value of:

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$$

Step-by-Step Solution

Define the function explicitly:

$$f(x, y) = \frac{\sqrt{x} + \sqrt{y}}{x + y}$$

Substitute $x \rightarrow xt$ and $y \rightarrow yt$ into the equation:

$$f(xt, yt) = \frac{\sqrt{xt} + \sqrt{yt}}{(xt) + (yt)}$$

Distribute the square roots in the numerator using $\sqrt{ab} = \sqrt{a}\sqrt{b}$:

$$f(xt, yt) = \frac{\sqrt{t}\sqrt{x} + \sqrt{t}\sqrt{y}}{xt + yt}$$

Factor out $\sqrt{t}$ (which is $t^{1/2}$) in the numerator and t (which is t^1) in the denominator:

$$f(xt, yt) = \frac{t^{1/2}(\sqrt{x} + \sqrt{y})}{t^1(x + y)}$$

Simplify the powers of t using the exponent rule $\frac{t^{1/2}}{t^1} = t^{1/2-1} = t^{-1/2}$:

$$f(xt, yt) = t^{-1/2} \left(\frac{\sqrt{x} + \sqrt{y}}{x + y} \right)$$

Substitute the original definition of u back into the equation:

$$f(xt, yt) = t^{-1/2}u$$

This confirms that u is a homogeneous function with a degree of $n = -\frac{1}{2}$.

By applying **Euler's Theorem** ($x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$):

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -\frac{1}{2}u$$

Final Answer

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -\frac{1}{2} \left(\frac{\sqrt{x} + \sqrt{y}}{x + y} \right)$$

Question 3

Problem Statement

If $z = \log(x^2 + y^2) + \frac{x^2 + y^2}{x + y} - 2 \log(x + y)$, prove that:

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{x^2 + y^2}{x + y}$$

Step-by-Step Solution

Step 1: Simplify the Function using Logarithmic Properties

Before evaluating homogeneity, group the logarithmic terms together using the standard power law $b \log(a) = \log(a^b)$:

$$z = \log(x^2 + y^2) - \log(x + y)^2 + \frac{x^2 + y^2}{x + y}$$

Apply the quotient rule for logarithms, $\log(m) - \log(n) = \log\left(\frac{m}{n}\right)$:

$$z = \log\left(\frac{x^2 + y^2}{(x + y)^2}\right) + \frac{x^2 + y^2}{x + y}$$

To make this expression easier to manage, split it into two independent substitute functions, u and v :

$$z = u + v$$

Where:

$$u = \log\left(\frac{x^2 + y^2}{(x + y)^2}\right)$$

$$v = \frac{x^2 + y^2}{x + y}$$

Step 2: Analyze Part 1 (u) for Homogeneity

Let the inner rational component of the logarithm be a distinct function:

$$f(x, y) = \frac{x^2 + y^2}{(x + y)^2} = \frac{x^2 + y^2}{x^2 + 2xy + y^2}$$

Substitute $x \rightarrow xt$ and $y \rightarrow yt$ into this inner algebraic component:

$$f(xt, yt) = \frac{(xt)^2 + (yt)^2}{(xt)^2 + 2(xt)(yt) + (yt)^2}$$

Factor out the common parameter t^2 from the numerator and denominator:

$$f(xt, yt) = \frac{t^2(x^2 + y^2)}{t^2(x^2 + 2xy + y^2)}$$

Cancel out the t^2 terms ($\frac{t^2}{t^2} = t^{2-2} = t^0$):

$$f(xt, yt) = t^0 \left(\frac{x^2 + y^2}{(x + y)^2} \right) = t^0 f(x, y)$$

Thus, the inner function is homogeneous of degree 0. Substitute this back into the full expression for u :

$$u(xt, yt) = \log\left(t^0 \cdot \frac{x^2 + y^2}{(x + y)^2}\right)$$

Because the scale factor has a power of 0, we can rewrite it using log properties ($\log(t^0 \cdot M) = \log(1 \cdot M)$):

$$u(xt, yt) = \log\left(\frac{x^2 + y^2}{(x + y)^2}\right) = u(x, y)$$

This means the overall function u itself behaves as a homogeneous function with a degree of $n_1 = 0$.

Applying **Euler's Theorem** to u :

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0 \cdot u = 0 \quad \text{--- (Equation 1)}$$

Step 3: Analyze Part 2 (v) for Homogeneity

Define the second function explicitly:

$$v(x, y) = \frac{x^2 + y^2}{x + y}$$

Substitute $x \rightarrow xt$ and $y \rightarrow yt$:

$$v(xt, yt) = \frac{(xt)^2 + (yt)^2}{(xt) + (yt)}$$

Factor out t^2 from the numerator and t^1 from the denominator:

$$v(xt, yt) = \frac{t^2(x^2 + y^2)}{t^1(x + y)}$$

Simplify the powers of t using the rule $\frac{t^2}{t^1} = t^{2-1} = t^1$:

$$v(xt, yt) = t^1 \left(\frac{x^2 + y^2}{x + y} \right) = t^1 v(x, y)$$

This confirms that v is a homogeneous function with a degree of $n_2 = 1$.

Applying **Euler's Theorem** to v :

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = 1 \cdot v = v \quad \text{--- (Equation 2)}$$

Step 4: Combine the Results using Linearity

Because differentiation is a linear operation, we can combine our separate differential operators back into our main target expression:

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = x \frac{\partial}{\partial x}(u + v) + y \frac{\partial}{\partial y}(u + v)$$

Distribute and re-group the terms into our known sub-equations:

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) + \left(x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} \right)$$

Substitute Equation 1 and Equation 2 directly into this combined structure:

$$\begin{aligned} x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} &= 0 + v \\ x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} &= v \end{aligned}$$

Substitute the original definition of v back into the finalized solution:

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{x^2 + y^2}{x + y} = \text{RHS}$$

Hence, proved.